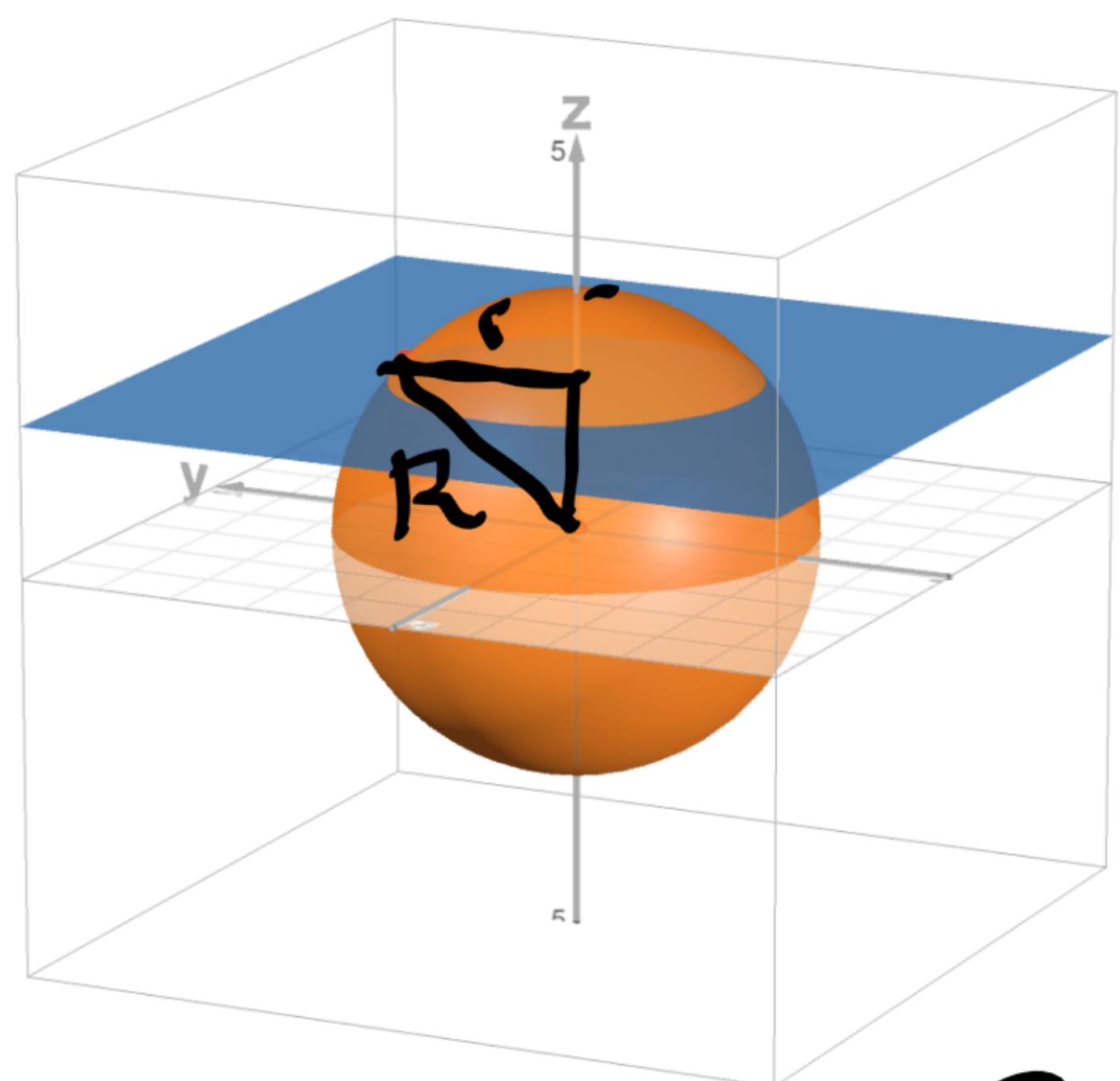
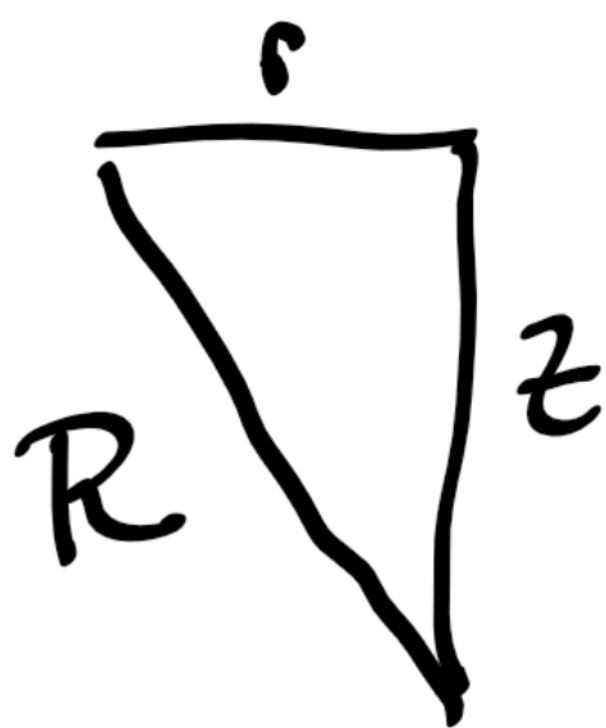




Radius R



$$R^2 = r^2 + z^2$$

$$r^2 = R^2 - z^2$$

$$r = \sqrt{R^2 - z^2}$$

$$\int_{-R}^R (\pi (R^2 - z^2)) dz = 2 \int_0^R \pi (R^2 - z^2) dz$$

$$= 2\pi \int_0^R (R^2 - z^2) dz$$

$$= 2\pi \left[\underbrace{\int_0^R R^2 dz}_{\text{}} - \int_0^R z^2 dz \right]$$

$$R^2 \int_0^R dz = R^2 z \Big|_0^R = R^3$$

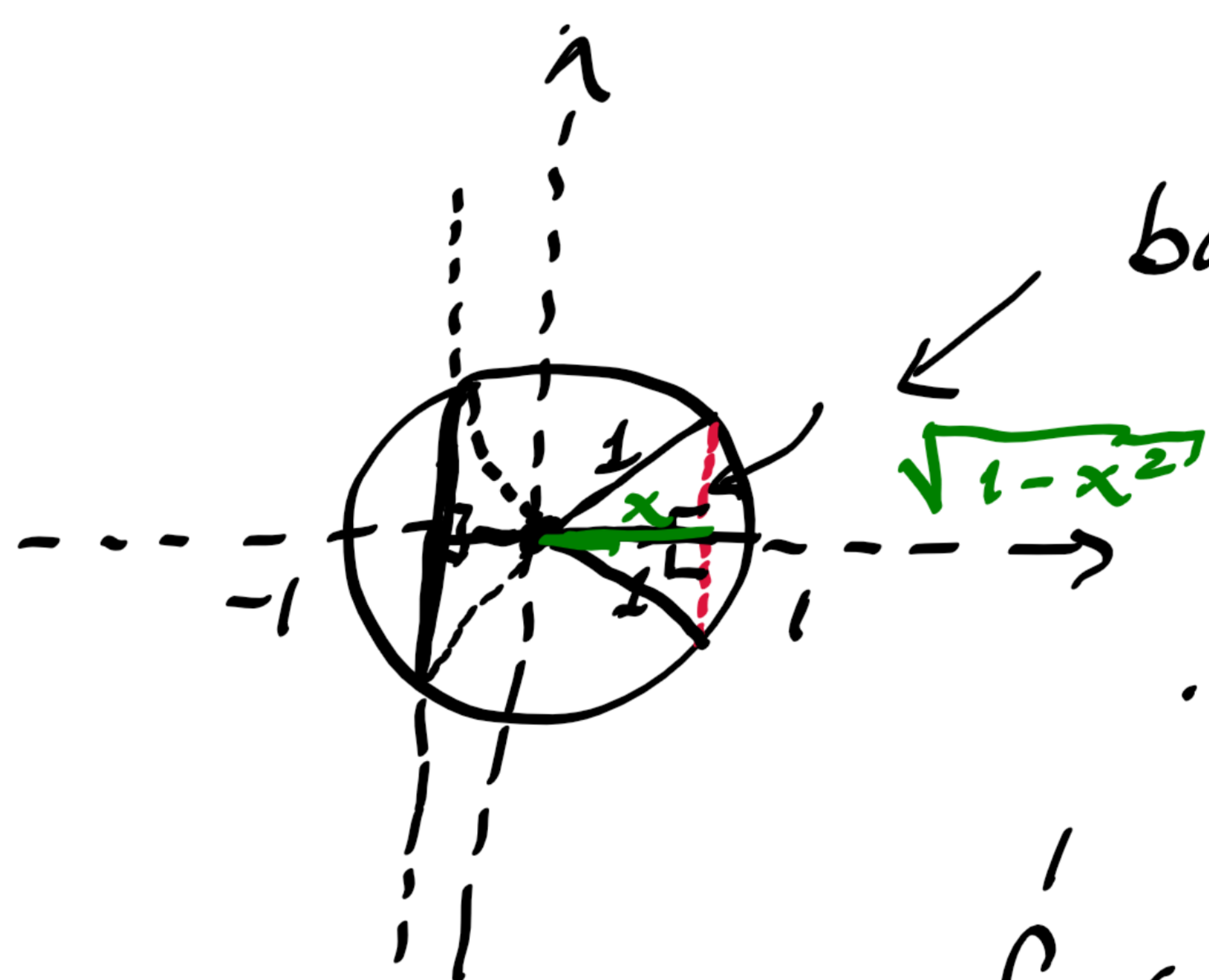
$$\int_0^R z^2 dz = \frac{1}{3} z^3 \Big|_0^R = \frac{1}{3} R^3$$

$$= 2\pi \left[R^3 - \frac{1}{3} R^3 \right] = 2\pi \left(\frac{2}{3} R^3 \right) = \frac{4\pi}{3} R^3$$

$$\int_a^b f(x) dx$$

$$= \underline{F(x)} \Big|_a^b$$

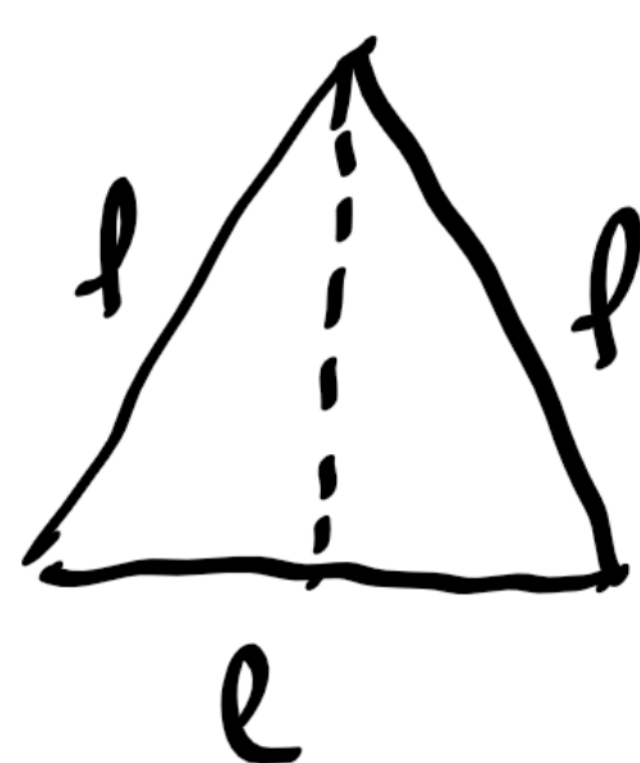
$$= F(b) - F(a)$$



base of volume.

Cross-sections perpendicular to the x-axis are equilateral triangles

$$\int_{-1}^1 (\text{Area of an equilateral triangle}) dx$$



$$\rightarrow \text{Area: } \frac{\sqrt{3}}{4} l^2$$

$$\int_{-1}^1 \frac{\sqrt{3}}{4} l^2 dx$$

$$\uparrow ? \quad 2\sqrt{1-x^2} = l(x)$$

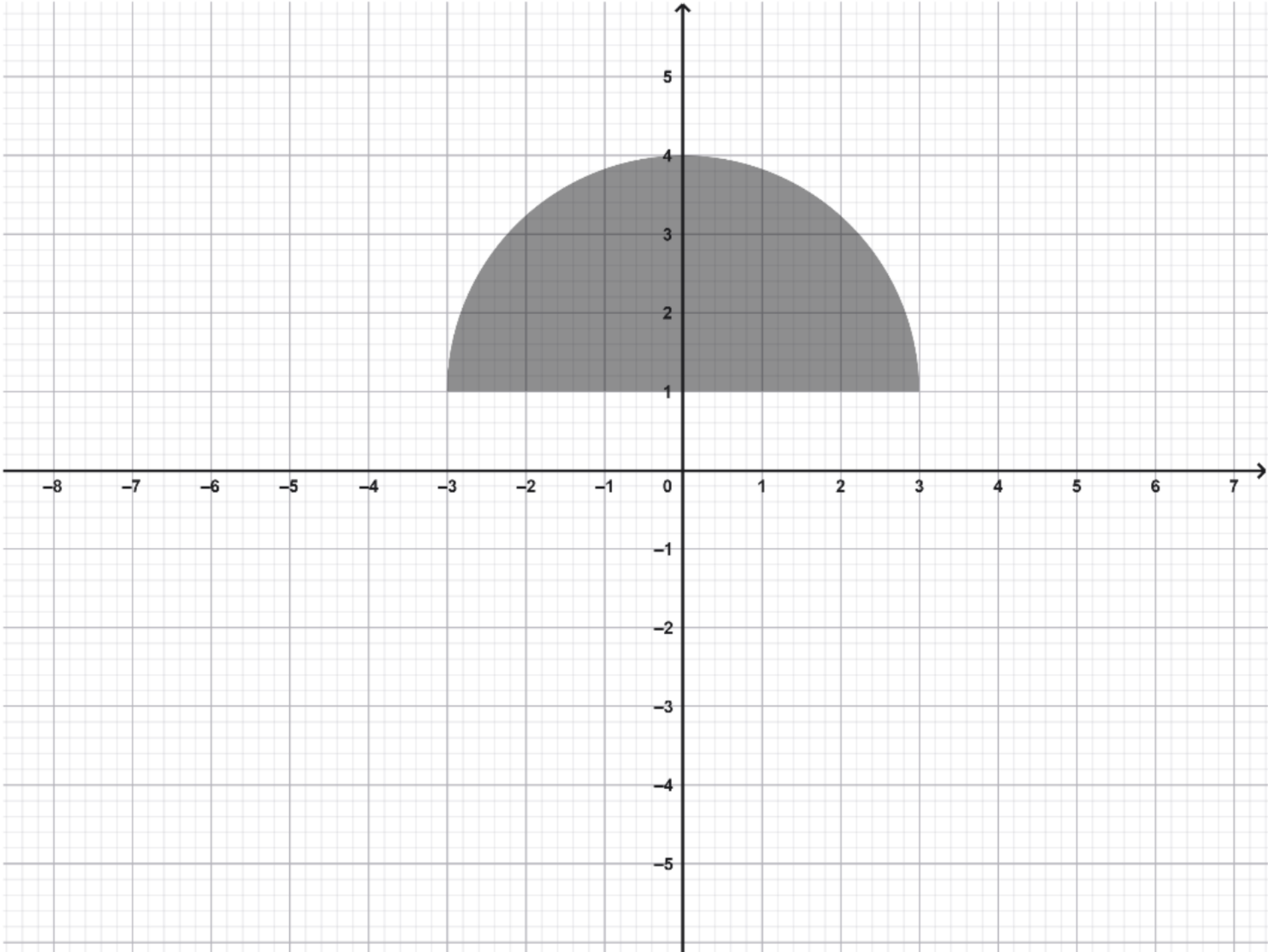
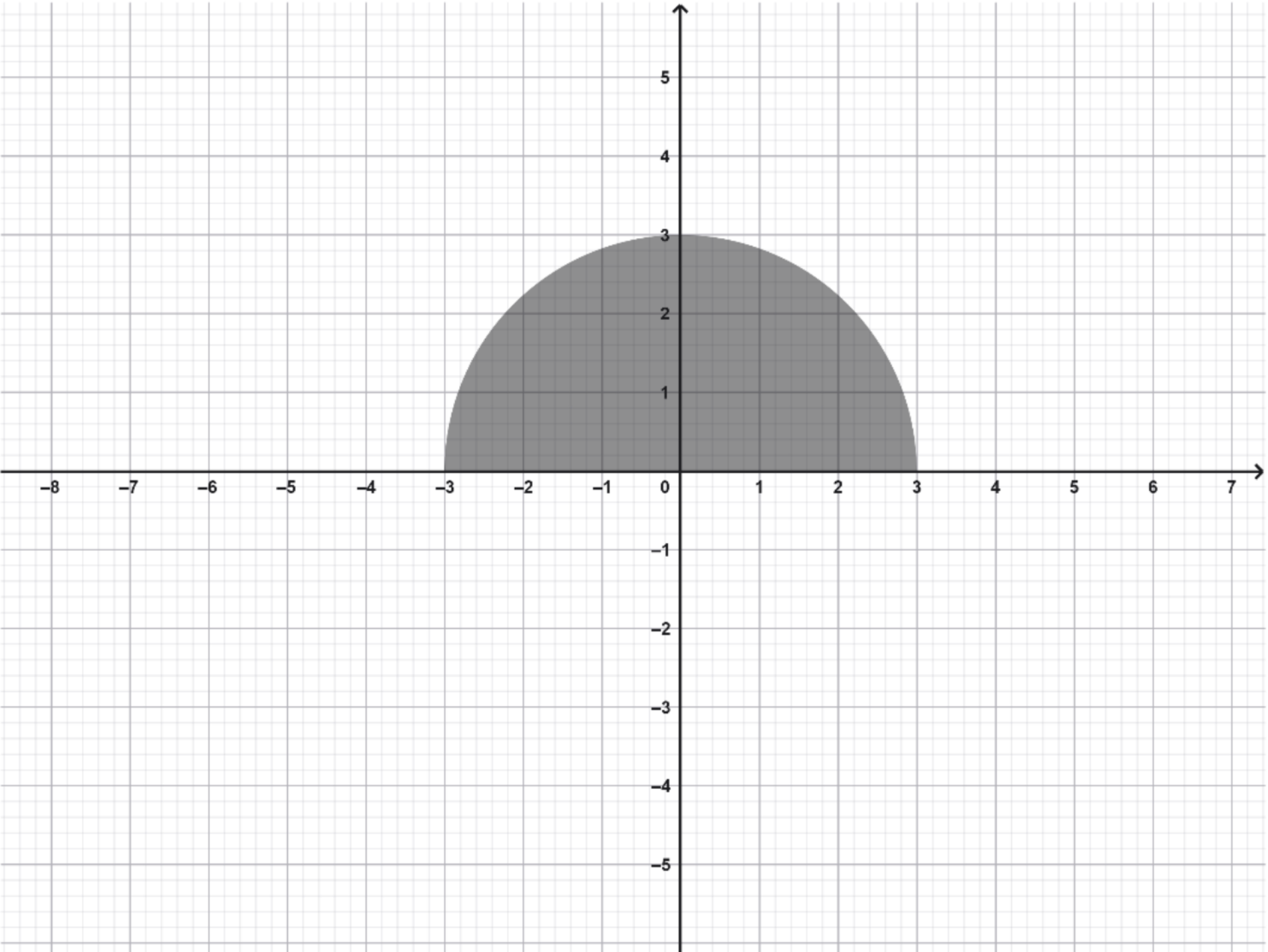
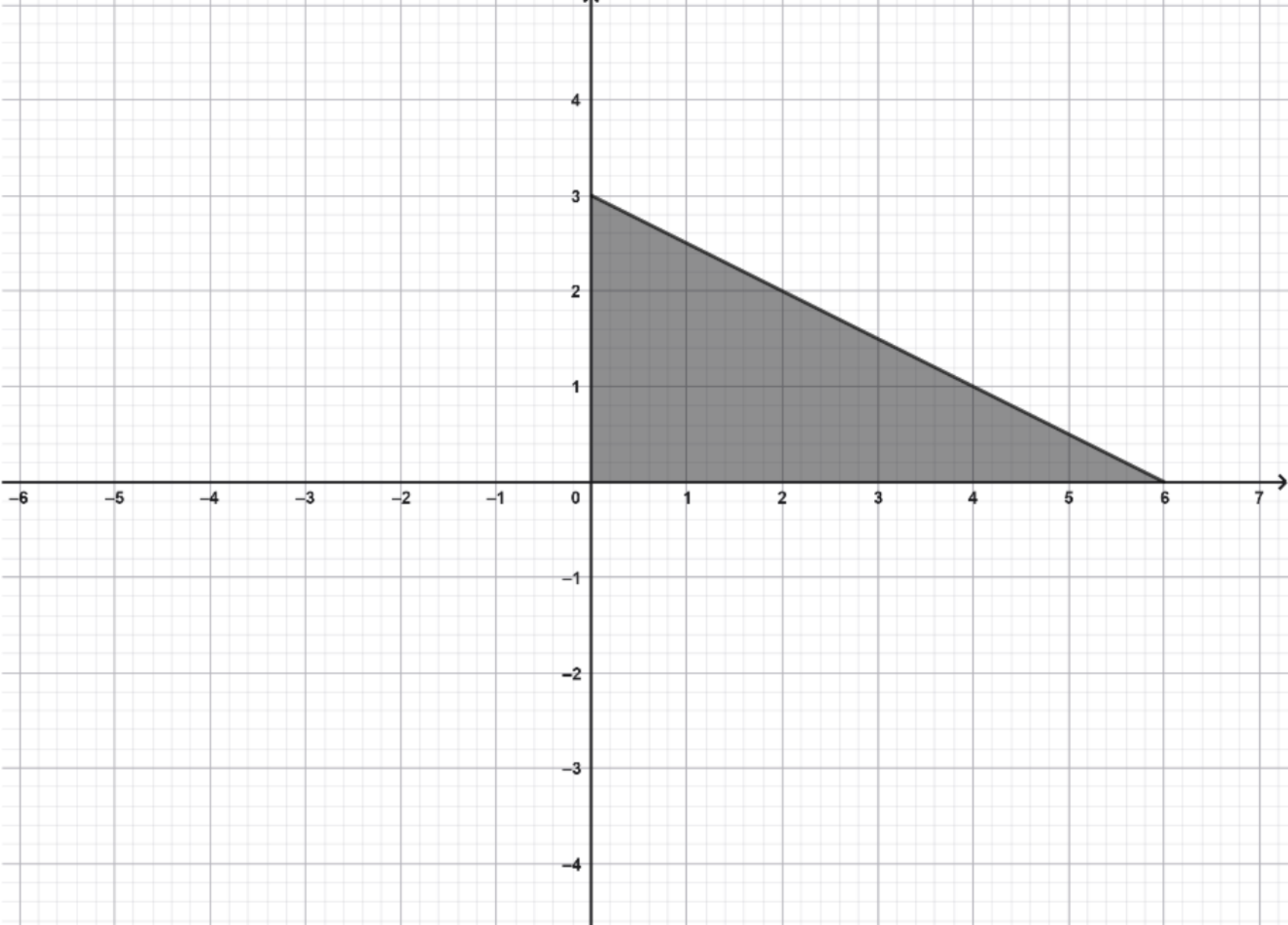
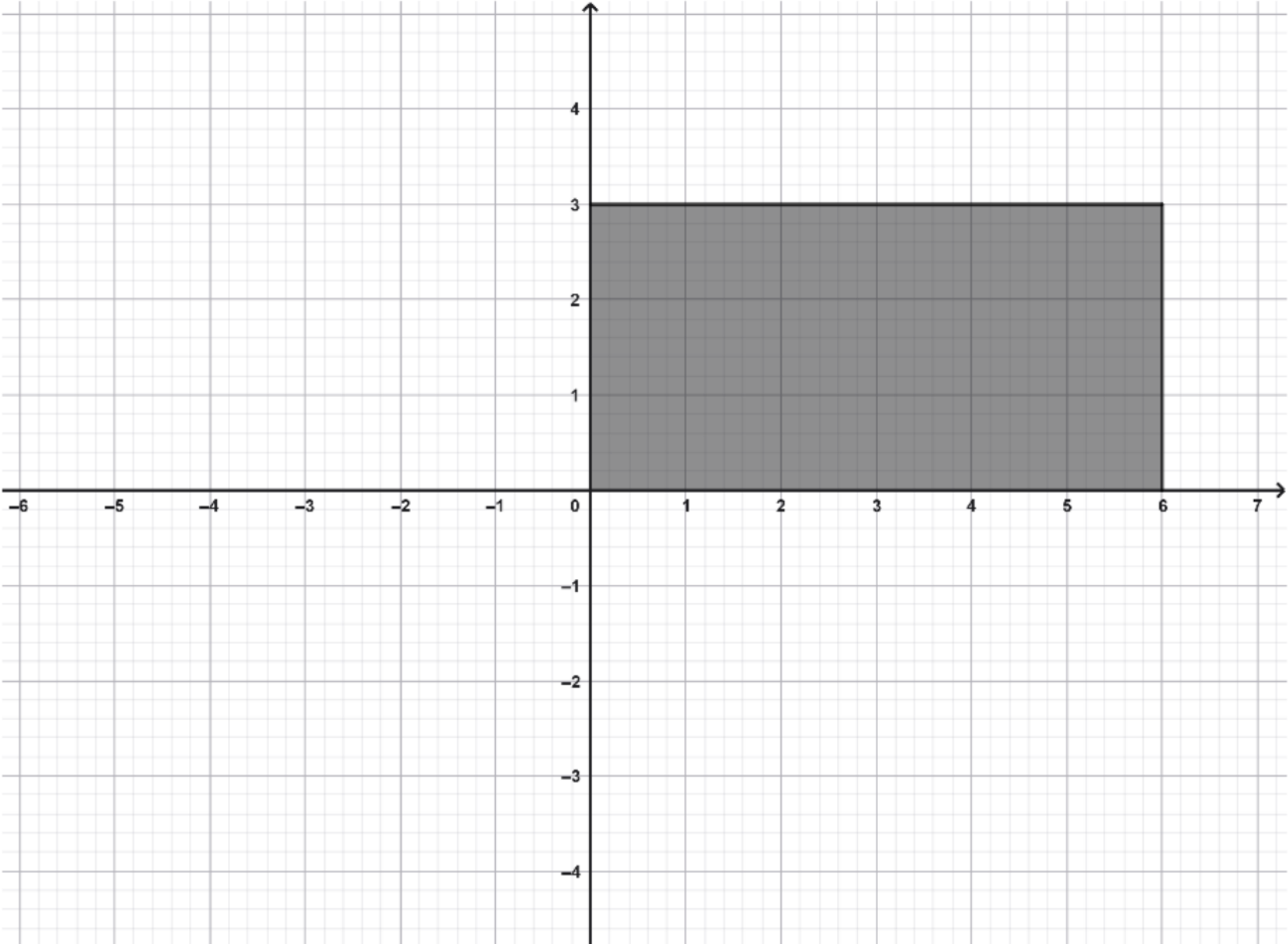
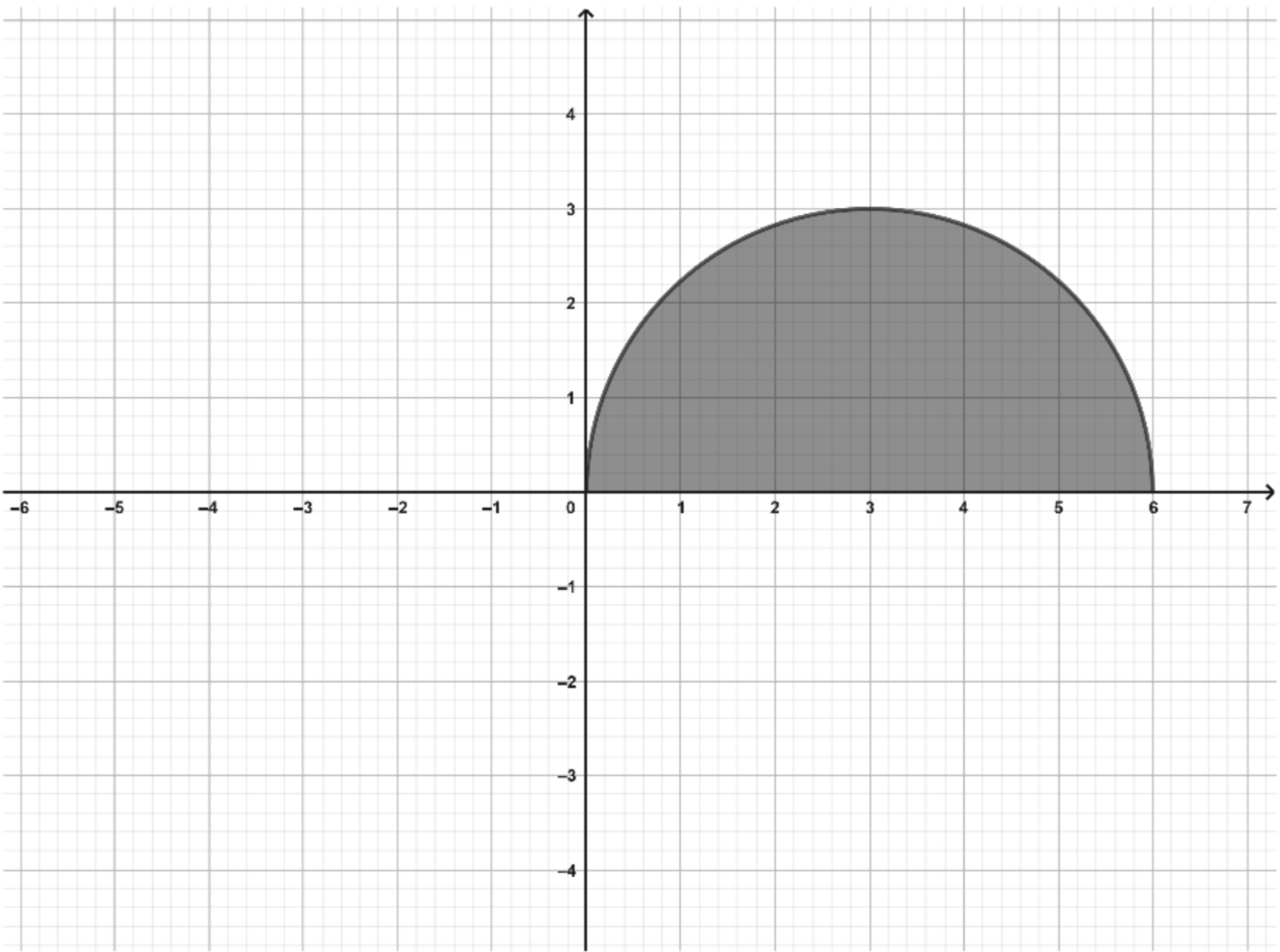
$$\int_{-1}^1 \frac{\sqrt{3}}{4} (2\sqrt{1-x^2})^2 dx$$

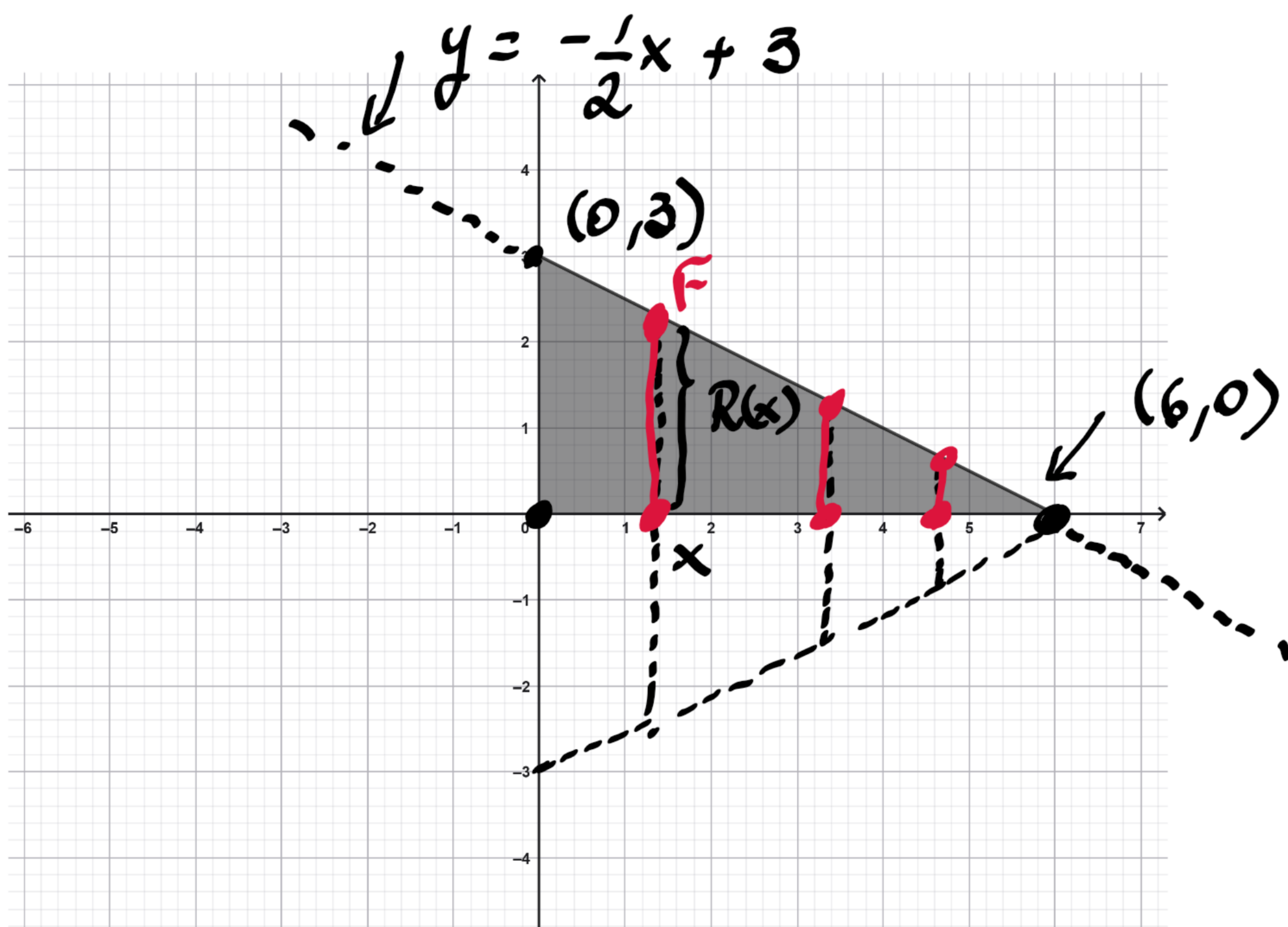
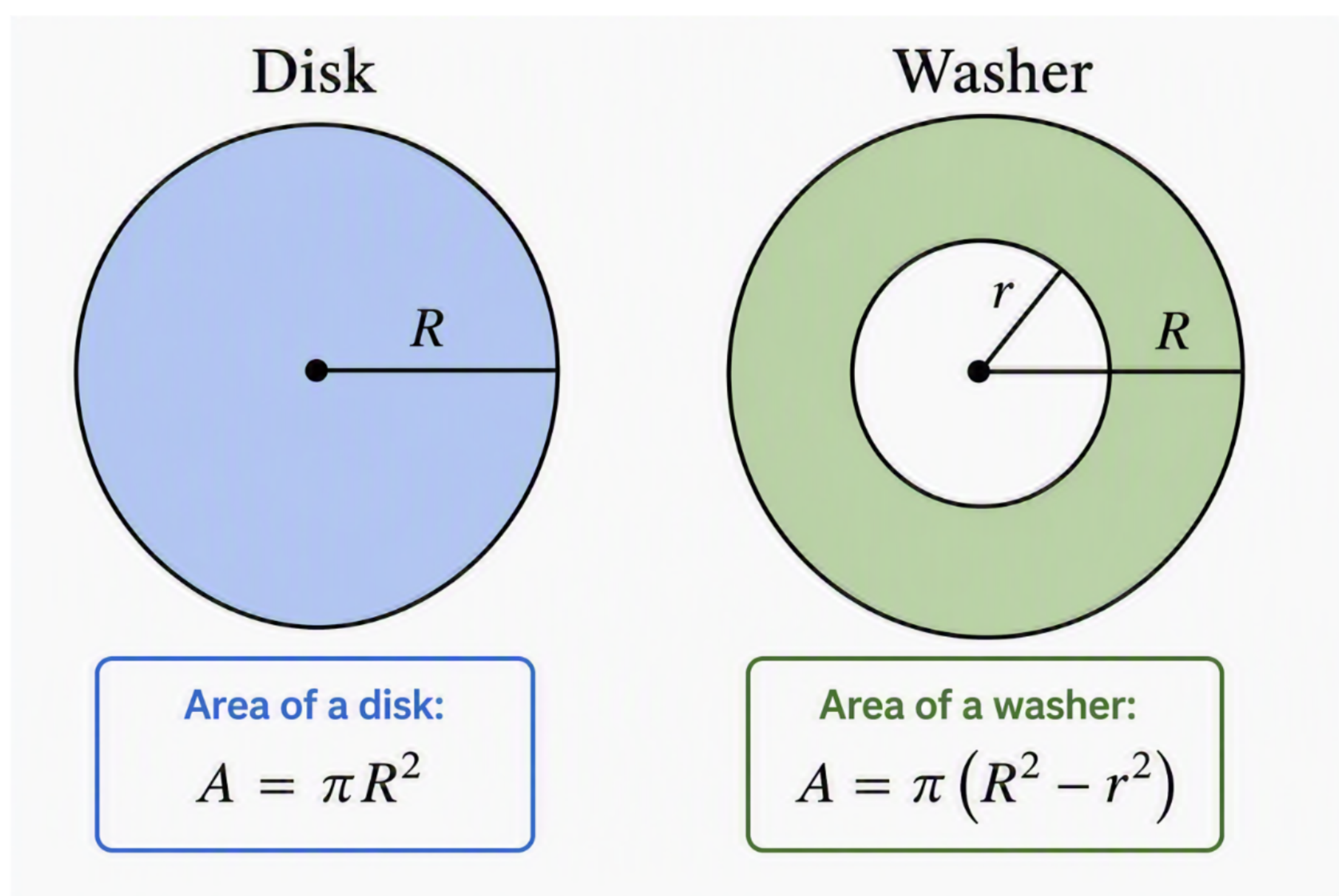
$$= \int_{-1}^1 \frac{\sqrt{3}}{4} \cdot 4 (1-x^2) dx = \sqrt{3} \int_{-1}^1 (1-x^2) dx = \underbrace{\sqrt{3} \frac{4}{3}}_{\frac{4}{3}}$$

$$\int_{-1}^1 (1-x^2) dx = x - \frac{x^3}{3} \Big|_{-1}^1$$

$$= 1 - \frac{1}{3} - \left((-1) - \frac{(-1)^3}{3} \right) = \frac{2}{3} - \left(-1 + \frac{1}{3} \right)$$

$$= \frac{2}{3} - \left(-\frac{2}{3} \right) = \frac{4}{3}$$





$$\int_0^6 \pi (R(x))^2 dx$$

$$= \int_0^6 \pi \left(3 - \frac{1}{2}x\right)^2 dx$$

$$R(x) = -\frac{1}{2}x + 3$$

$$= \pi \int_0^6 \left(3 - \frac{1}{2}x\right)^2 dx = \pi \int_3^0 u^2 (-2 du) = -2\pi \int_3^0 u^2 du$$

$$u = 3 - \frac{1}{2}x$$

$$du = -\frac{1}{2}dx$$

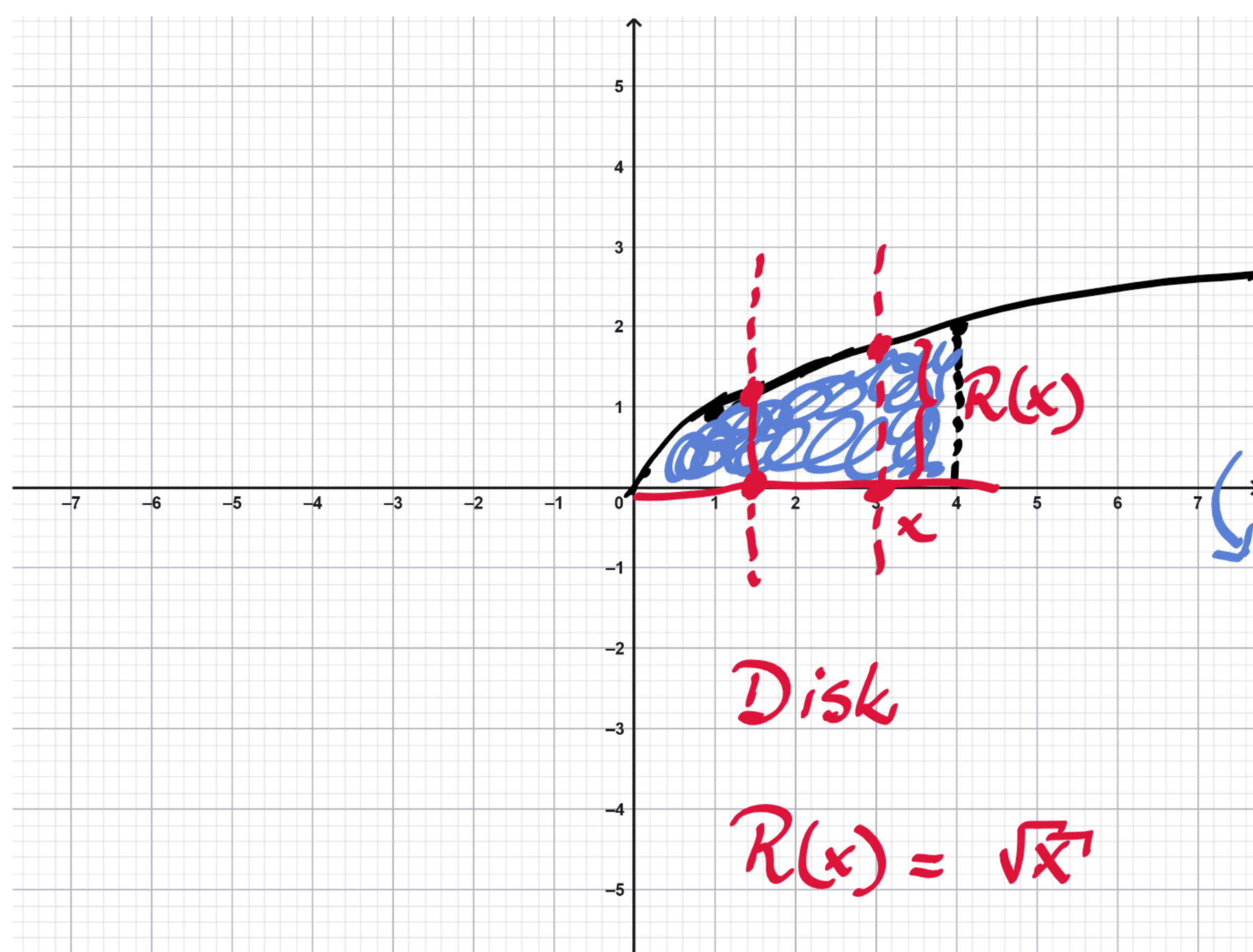
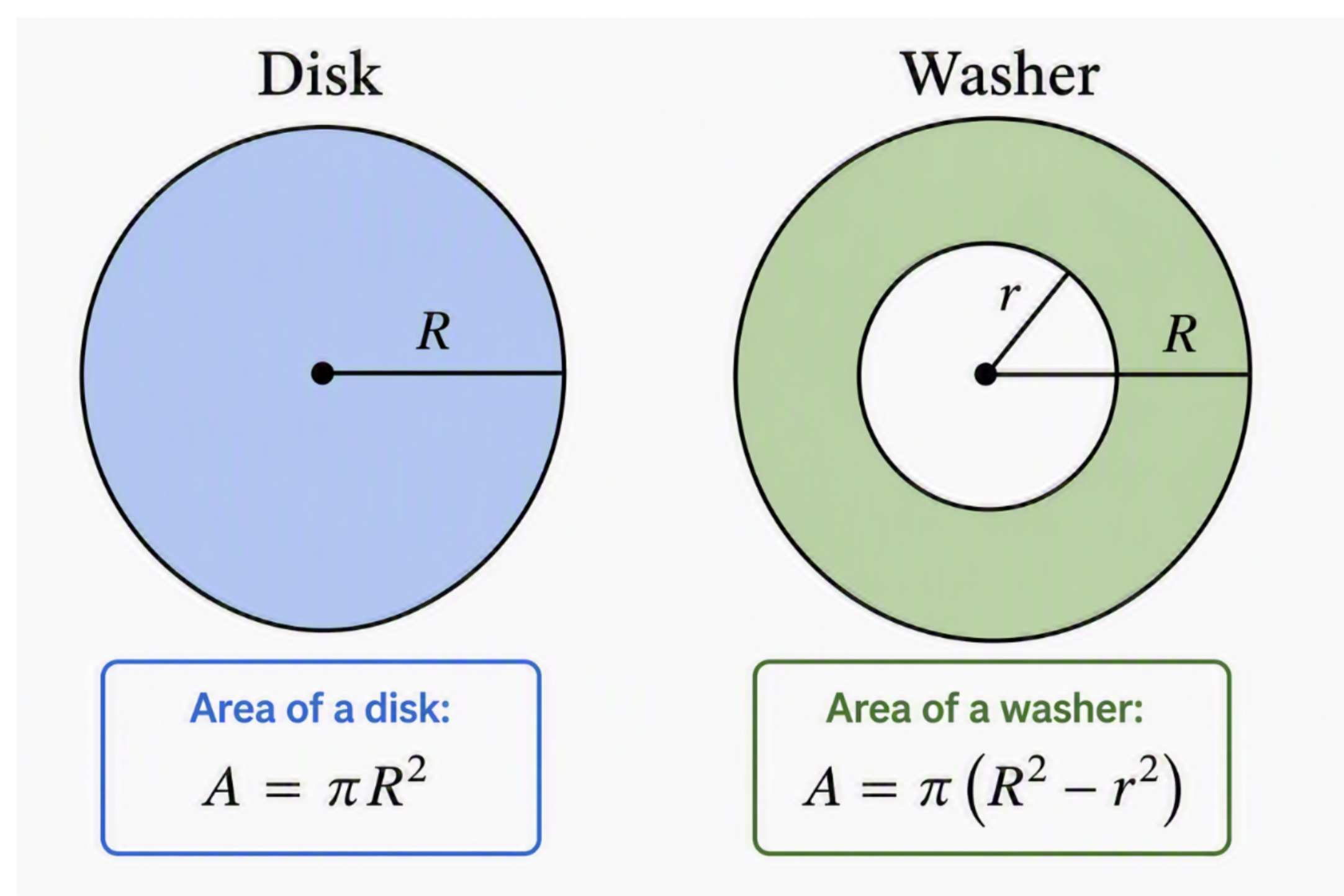
$$= 2\pi \int_0^3 u^2 du$$

$$= 2\pi \left. \frac{u^3}{3} \right|_0^3$$

$$= 2\pi \left(\frac{3^3}{3} \right) = 18\pi$$

$$-2\pi \left. \frac{u^3}{3} \right|_3^0 = -2\pi \left(0 - \frac{27}{3} \right)$$

$$= 18\pi$$

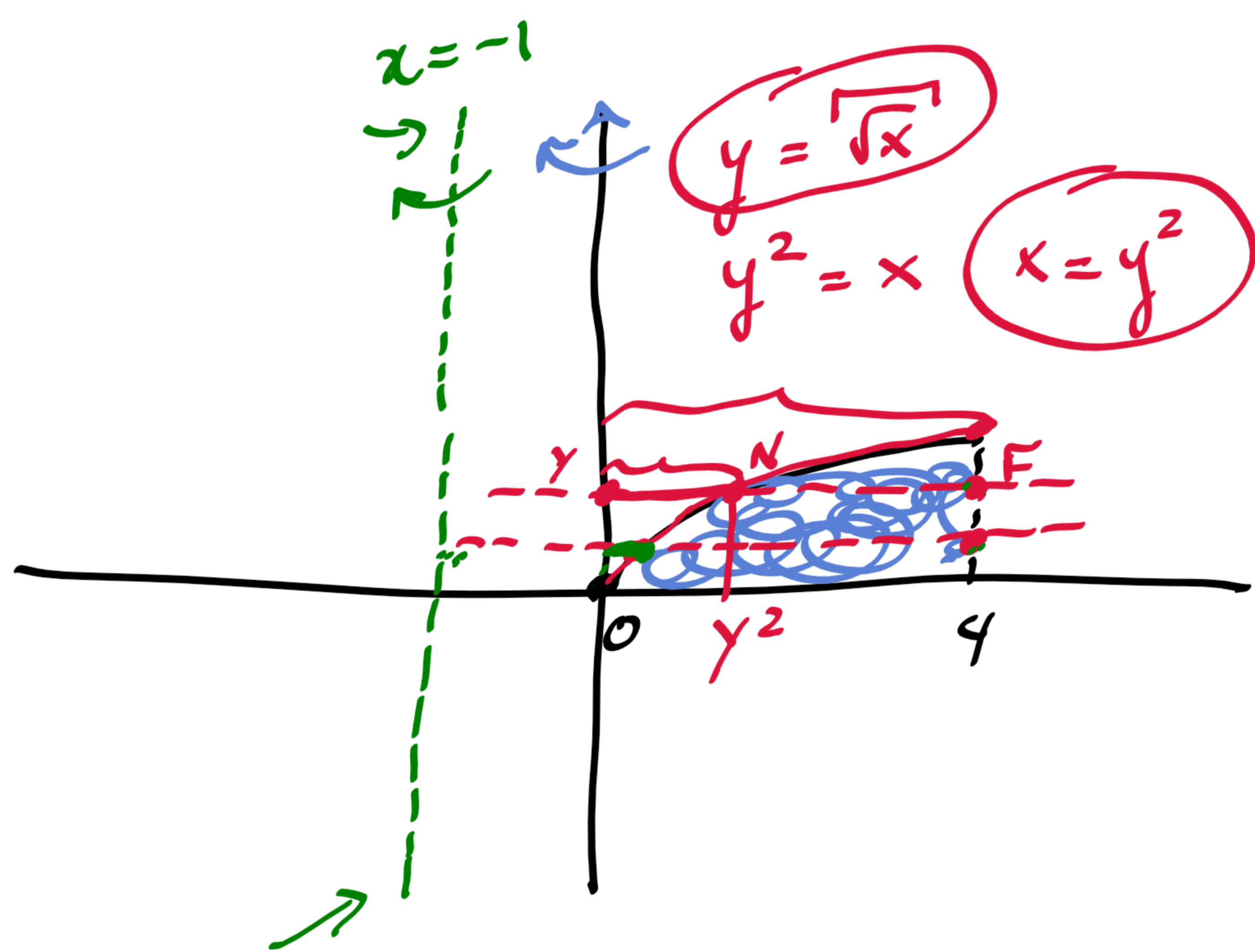


Rotate region bounded
 by $y = \sqrt{x}$
 x-axis

$x=0$, $x=4$
 about x-axis.

$$\int_0^4 \pi (\sqrt{x})^2 dx = \pi \int_0^4 x dx$$

$$= \pi \frac{x^2}{2} \Big|_0^4 = 8\pi$$



$$\int_0^2 \pi (5^2 - (y^2 + 1)^2) dy$$

$$\int_0^2 \pi (4^2 - (y^2)^2) dy$$

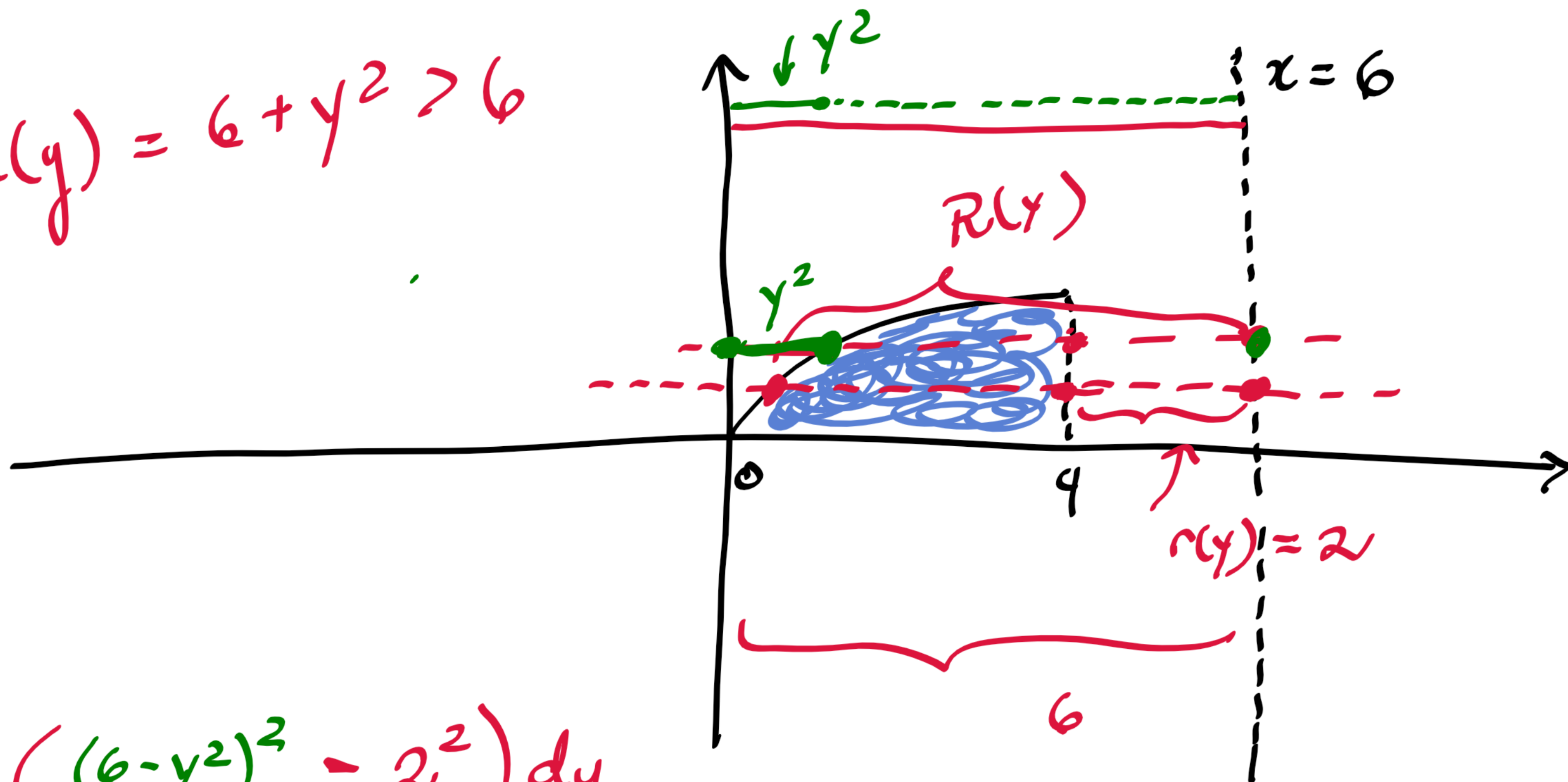
$$= \pi \int_0^2 (16 - y^4) dy$$

$$= \pi \left(16y - \frac{y^5}{5} \Big|_0^2 \right)$$

$$= \pi \left(32 - \frac{32}{5} \right)$$

$$= \pi \frac{4}{5} \cdot 32 = \frac{128}{5} \pi$$

$$R(y) = 6 + y^2 > 6$$



$$\int_0^2 \pi \left((6-y^2)^2 - 2^2 \right) dy$$